Sample Paper 16

Class - 12th Exam - 2025 - 26

Mathematics (Code-041)

Time : 3 Hours **Max. Marks : 80**

General Instructions :

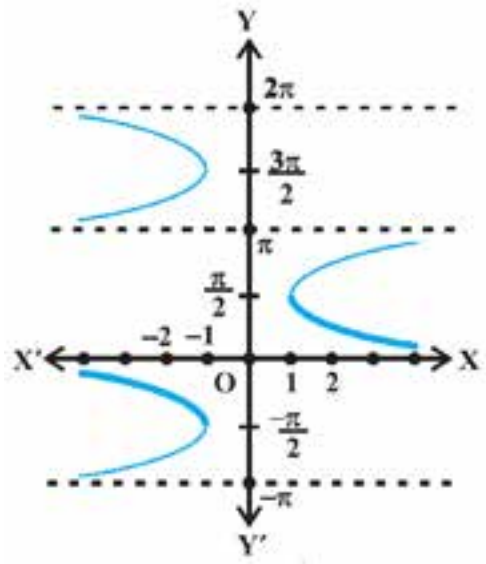
Read the following instructions very carefully and strictly follow them :

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, B, C, D and E.
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. In Section B, Questions no. 21 to 25 are Very Short Answer (VSA)-type questions, carrying 2 marks each.
5. In Section C, Questions no. 26 to 31 are Short Answer (SA)-type questions, carrying 3 marks each.
6. In Section D, Questions no. 32 to 35 are Long Answer (LA)-type questions, carrying 5 marks each.
7. In Section E, Questions no. 36 to 38 are Case study-based questions, carrying 4 marks each.
8. There is no overall choice. However, an internal choice has been provided some questions.
9. Use of calculators is not allowed.

SECTION - A

Section A consists of 20 questions of 1 mark each.

1. What is the range of the function $y = \csc^{-1} x$ as represented in the graph?



(a) $[-\pi/2, \pi/2]$ (b) $[0, \pi] \setminus \{\pi/2\}$

(c) $(-\pi, \pi)$ (d) $(-\infty, \infty)$

2. The solution of the differential equation $2x\frac{dy}{dx} - y = 3$ represents

(a) straight line (b) circle

(c) parabola (d) ellipse

3. The solution of the differential equation $\frac{dy}{dx} + \frac{y}{x} = x^2$ is

(a) $x + y = \frac{x^2}{2} + c$ (b) $x - y = \frac{1}{3}x^3 + c$

(c) $xy = \frac{1}{4}x^4 + c$ (d) $y - x = \frac{1}{4}x^4 + c$

4. Let D, E, F are the mid points of sides BC, CA, AB respectively of $\triangle ABC$. Which of the following is true?

(a) $\overrightarrow{AB} = 2\overrightarrow{ED}$ (b) $\overrightarrow{AB} = 2\overrightarrow{DE}$

(c) $\overrightarrow{AB} = \overrightarrow{ED}$ (d) $\overrightarrow{AB} = 2\overrightarrow{DF}$

5. The points (5, 2, 4), $(6, -1, 2)$ and $(8, -7, k)$ are collinear, if k is equal to

(a) -2 (b) 2

(c) 3 (d) -1

6. If A and B are two independent events such that $P(A) = \frac{1}{2}$ and $P(B) = \frac{1}{3}$, then P (neither A nor B) is equal to

(a) $\frac{2}{3}$ (b) $\frac{1}{6}$

(c) $\frac{5}{6}$ (d) $\frac{1}{3}$

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7. Find the values of x, y and z from the following equations $\begin{bmatrix} 4 & x-z \\ 2+y & xz \end{bmatrix} = \begin{bmatrix} 4 & 3 \\ -1 & 10 \end{bmatrix}$.]

(a) $x=-5,\ y=3,\ z=2$

(b) $x=5,\ y=-3,\ z=2$

(c) $x=5,\ y=3,\ z=-2$

(d) $x=5,\ y=-3,\ z=-2$

8. If $y = a\log|x| + bx^2 + x$ has its extreme value at $x=-1$ and $x=2$, then

(a) $a=2, b=-1$ (b) $a=2, b=-\frac{1}{2}$

(c) $a=-2, b=\frac{1}{2}$ (d) $a=-2, b=-\frac{1}{2}$

9. The point of inflexion for the curve $y=x^{5/3}$ is

(a) $(1,1)$ (b) $(0,0)$

(c) $(1,0)$ (d) $(0,1)$

10. If $\vec{a}$ and $\vec{b}$ are two unit vectors inclined to x-axis at angles $30°$ and $120°$, then $|\vec{a}\times\vec{b}|$ equals

(a) $\sqrt{\frac{2}{3}}$ (b) $\sqrt{2}$

(c) $\sqrt{3}$ (d) 2

11. $\int_0^{\pi/2} \left|\cos\left(\frac{x}{2}\right)\right| dx$ is equal to

(a) 1 (b) -2

(c) $\sqrt{2}$ (d) 0

12. The point of discontinuous of $\tan x$ are

(a) $n\pi,\ n\in I$ (b) $2n\pi,\ n\in I$

(c) $(2n+1)\frac{\pi}{2},\ n\in I$ (d) None of these

13. The area bounded by the curve $y=\frac{1}{2}x^2$, the X-axis and the ordinate $x=2$ is

(a) $\frac{1}{3}$ sq unit (b) $\frac{2}{3}$ sq unit

(c) 1 sq unit (d) $\frac{4}{3}$ sq unit

14. If A and B are two equivalence relations defined on set C, then

(a) $A\cap B$ is an equivalence relation

(b) $A\cap B$ is not an equivalence relation

(c) $A\cap B$ is an equivalence relation

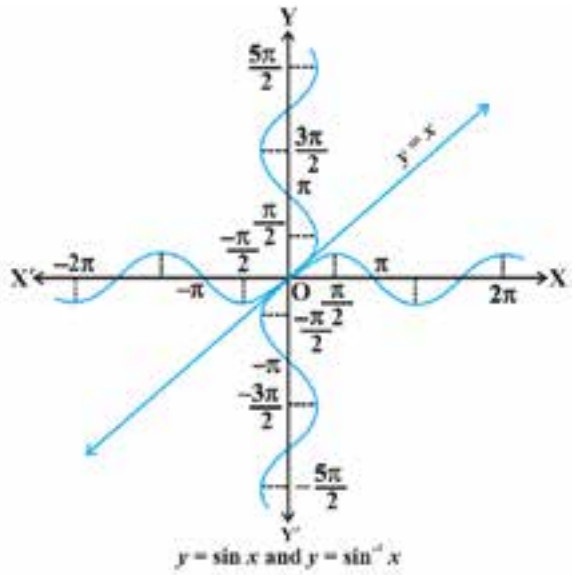
(d) $A\cap B$ is not an equivalence relation

15. The range of the function $f(x)=x^2+2x+2$ is

(a) $(1,\infty)$ (b) $(2,\infty)$

(c) $(0,\infty)$ (d) $[1,\infty)$

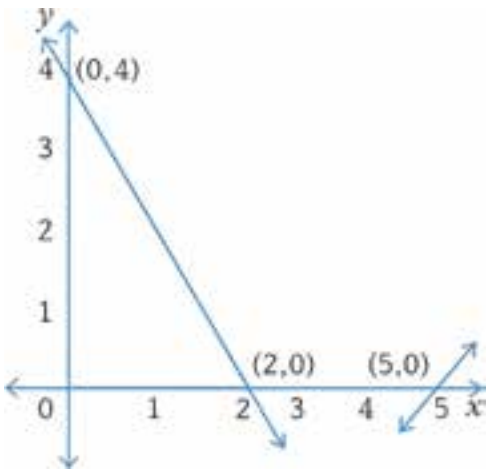
16. What is the period of the function $y=\tan x$ as seen in the graph?



(a) π (b) 2π

(c) $\pi/2$ (d) Infinite

17. For the LPP shown in the figure, the objective is Minimise $Z=3x+2y$. From the graph, the minimum value of Z is obtained at the corner point (2,0). The optimal solution of this LPP:



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(a) Does not exist as the feasible region is unbounded.

(b) Does not exist as the inequality 3x+2y<6 does not have any point in common with the feasible region.

(c) Exists as the inequality 3x+2y>6 has infinitely many points in common with the feasible region.

(d) Exists as the inequality 3x+2y<6 does not have any point in common with the feasible region.

18. The solution of the equation $(x^2 - xy)\,dy = (xy + y^2)\,dx$ is

(a) $xy = ce^{-y/x}$ (b) $xy = ce^{-x/y}$

(c) $yx^2 = ce^{1/x}$ (d) none of these

19. **Assertion:** if $2P(A) = P(B) = \frac{5}{13}$ and $P\left(\frac{A}{B}\right) = \frac{2}{5}$, then $P(A \cup B)$ is $\frac{11}{26}$

Reason: E_1 and E_2 are two events. then

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cup E_2)}{P(E_2)},\ 0 < P(E_2) \le 1$$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

20. Consider, if $u = f(n)$, $v = g(x)$, then the derivative of f with respect to g is $\frac{du}{dv} = \frac{du/dx}{dv/dx}$.

Assertion: Derivative of $\sin^{-1}\left(\frac{2x}{1+x^2}\right)$ with respect to $\cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ is 1 for $0 < x < 1$

Reason: $\sin^{-1}\left(\frac{2x}{1+x^2}\right) \ne \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right)$ for $-1 \le 1x \le 1$

(a) Assertion is true, Reason is true; Reason is a correct explanation for Assertion.

(b) Assertion is true, Reason is true; Reason is not a correct explanation for Assertion.

(c) Assertion is true; Reason is false.

(d) Assertion is false; Reason is true.

SECTION - B

This section comprises of very short answer type-questions (VSA) of 2 marks each.

21. If $|\vec{a}| = 8$, $|\vec{b}| = 3$ and $|\vec{a} \times \vec{b}| = 12$, find the angle between $\vec{a}$ and $\vec{b}$.

OR

Write the value of λ, so that the vectors $\vec{a} = 2\hat{i} + \lambda\hat{j} + \hat{k}$ and $\vec{b} = \hat{i} - 2\hat{j} + 3\hat{k}$ are perpendicular to each other.

22. The equation of a line is

$$5x - 3 = 15y + 7 = 3 = 3 - 10z$$

Write the direction cosines of the line.

23. Find the vector equation of the line which passes through the point (3,4,5) and is parallel to the vector $2\hat{i} + 2\hat{j} - 3\hat{k}$.

24. Show that the relation R on IR defined as $R = \{(a, b): (a \le b)\}$, is reflexive and transitive but not symmetric.

25. Write the value of $\int \frac{2 - 3\sin x}{\cos^2 x}\,dx$.

OR

Write the value of $\int \sec x(\sec x + \tan x)\,dx$.

SECTION - C

This section comprises of short answer type questions (SA) of 3 marks each.

26. If $y = (\sin x)^x + \sin^{-1}\sqrt{x}$, then find $\frac{dy}{dx}$.

27. If θ is the angle between two vectors $\hat{i} - 2\hat{j} + 3\hat{k}$ and $3\hat{i} - 2\hat{j} + \hat{k}$, find $\sin\theta$.

28. Evaluate $\int_0^{\pi} |x^3 - x|\,dx$.

OR

Evaluate $\int_{-2}^{2} \frac{x^2}{1+5^x}\,dx$.

29. Prove that $\tan\left\{\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{a}{b}\right\}$

$$\tan\left\{\frac{\pi}{4} + \frac{1}{2}\cos^{-1}\frac{a}{b}\right\} + \tan\left\{\frac{\pi}{4} - \frac{1}{2}\cos^{-1}\left(\frac{a}{b}\right)\right\} = \frac{2b}{a}$$

30. If $\begin{vmatrix} 3x & 5 \\ 8 & x \end{vmatrix} = \begin{vmatrix} 6 & -2 \\ 7 & 3 \end{vmatrix}$, then write the value of x.

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31. Show that $y = \log(1+x) - \frac{2x}{2+x}$, $x > -1$ is an increasing function of x, throughout its domain.

OR

Find the intervals in which the function given by $f(x) = \sin x + \cos x$, $0 \le x \le 2\pi$ is

(i) increasing

(ii) decreasing.

SECTION - D

This section comprises of long answer-type questions (LA) of 5 marks each.

32. Maximize $Z = 3x + 4y$, subject to the constraints; $x + y \le 4$, $x \ge 0$, $y \ge 0$.

OR

Minimize $Z = -3x + 4y$ subject to the constraints

$$x + 2y \le 8,$$

$$3x + 2y \le 12,$$

$$x \ge 0,\ y \ge 0$$

33. Evaluate $\int_0^{\pi/2} \frac{x \sin x \cos x}{\sin^4 x + \cos^4 x} dx$.

34. Solve the following differential equation $\operatorname{cosec} x \log|y| \frac{dy}{dx} + x^2 y^2 = 0$.

OR

Solve the following differential equation.

$$x \cos\left(\frac{y}{x}\right)\frac{dy}{dx} = y\cos\left(\frac{y}{x}\right) + x;\ x \neq 0$$

35. Find the values of p, so that the lines

$$l_1 : \frac{1-x}{3} = \frac{7y-14}{p} = \frac{z-3}{2} \text{ and}$$

$$l_2 : \frac{7-7x}{3p} = \frac{y-5}{1} = \frac{6-z}{5}$$

are perpendicular to each other. Also, find the equation of a line passing through a point $(3, 2, -4)$ and parallel to line l_1.

SECTION - E

Case study based questions are compulsory.

36. Publishing is the activity of making information, literature, music, software and other content available to the public for sale or for free. Traditionally, the term refers to the creation and distribution of printed works, such as books, newspapers, and magazines.



NODIA Press is a such publishing house having two branch at Jaipur. In each branch there are three offices. In each office, there are 2 peons, 5 clerks and 3 typists. In one office of a branch, 5 salesmen are also working. In each office of other branch 2 head-clerks are also working. Using matrix notations find:

(i) The total number of posts of each kind in all the offices taken together in each branch.

(ii) The total number of posts of each kind in all the offices taken together from both branches.

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37. Mahesh runs a form processing agency. He collect form from different office and then extract data and record data on computer. In his office three employees Vikas, Sarita and Ishaan process incoming copies of a form. Vikas process 50% of the forms. Sarita processes 20% and Ishaan the remaining 30% of the forms. Vikas has an error rate of 0.06, Sarita has an error rate of 0.04 and Ishaan has an error rate of 0.03.



Based on the above information answer the following questions.

(i) The total probability of committing an error in processing the form.

(ii) The manager of the company wants to do a quality check. During inspection he selects a form at random from the days output of processed forms. If the form selected at random has an error, the probability that the form is not processed by Vikas.

38. RK Verma is production analysts of a ready-made garment company. He has to maximize the profit of company using data available. He find that $P(x) = -6x^2 + 120x + 25000$ (in Rupee) is the total profit function of a company where x denotes the production of the company.



Based on the above information, answer the following questions.

(i) Find the profit of the company, when the production is 3 units.

(ii) Find P' (5)

(iii) Find the interval in which the profit is strictly increasing.

OR

(iv) Find the production, when the profit is maximum.

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